RACE
Regular Analysis Through Continuous Exercise

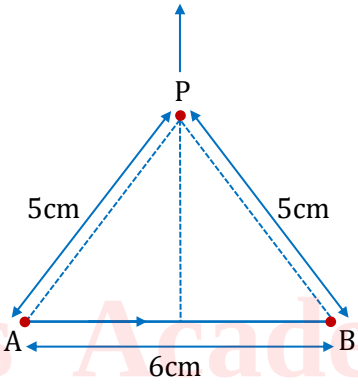
Magnetic Effects of Current and Magnetism : RACE-01

1. A regular polygon of 6 sides is formed by bending a wire of length 4 π meter. If an electric current of $4\pi\sqrt{3}$ A is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x \times 10^{-7}$ T. The value of x is ___.

PMG216

2. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (See figure)

($\mu_0 = 4\pi \times 10^{-7}$ $N\text{-}A^{-2}$)



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(A) 3.0×10^{-5} T (B) 2.5×10^{-5} T (C) 2.0×10^{-5} T (D) 1.5×10^{-5} T

PMG217

3. Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is :

(A) $250\sqrt{3}$ (B) $5\sqrt{3}$ (C) $500\sqrt{3}$ (D) $50\sqrt{3}$

PMG218

4. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil :

(A) $\frac{\mu_0 IN}{2(b-a)} log_e\left(\frac{b}{a}\right)$ (B) $\frac{\mu_0 I}{8}\left[\frac{a+b}{a-b}\right]$ (C) $\frac{\mu_0 I}{4(a-b)}\left[\frac{1}{a}-\frac{1}{b}\right]$ (D) $\frac{\mu_0 I}{8}\left(\frac{a-b}{a+b}\right)$

PMG219

5. The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is :

(A) $\frac{5}{2}$ (B) $\frac{25}{4}$ (C) $\frac{5}{4}$ (D) $\frac{25}{2}$

PMG220

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6. The magnetic field at the center of current carrying circular loop is B_1. The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2. The value of B_1/B_2 will be

(A) 9 : 4 (B) 12 : $\sqrt{5}$ (C) 8 : 1 (D) 5 : $\sqrt{3}$

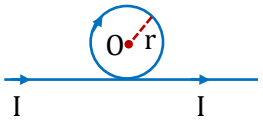
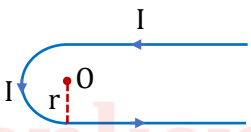
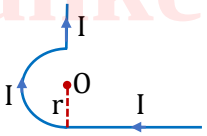
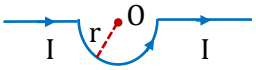
PMG221

7. A closely wounded circular coil of radius 5 *cm* produces a magnetic field of 37.68 x 10^{-4} *T* at its center. The current through the coil is _____ *A*.

[Given, number of turns in the coil is 100 and π = 3.14]

PMG222

8. Math List I with List II

List – I (Current configuration)		**List – II (Magnetic field at point O)**	
A		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C		III.	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D		IV.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the option given below:

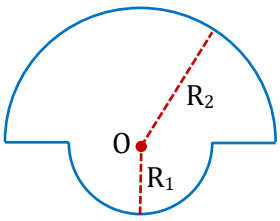
(A) A-III, B-IV, C-I, D-II (B) A-I, B-III, C-IV, D-II

(C) A-III, B-I, C-IV, D-II (D) A-II, B-I, C-IV, D-III

PMG223

9. The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi$ *m* and $R_2 = 4\pi$ *m* carrying current $I = 4A$ as per figure given below is $\alpha \times 10^{-7}$ *T*. The value of α is _____.

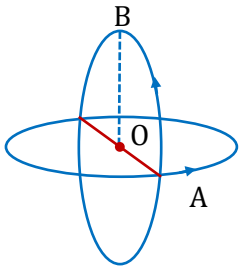
(Centre O is common for all segments)



PMG224

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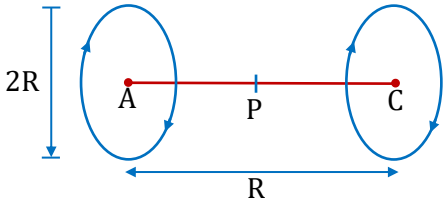
10. Two insulated circular loop *A* and *B* radius '*a*' carrying a current of '*I*' in the anti clockwise direction as shown in figure. The magnitude of the magnetic induction at the centre will be :



(A) $\frac{\sqrt{2}\mu_0 I}{a}$ (B) $\frac{\mu_0 I}{2a}$ (C) $\frac{\mu_0 I}{\sqrt{2}a}$ (D) $\frac{2\mu_0 I}{a}$

PMG225

11. A Helmholtz coil has a pair of loops, each with *N* turns and radius *R*. They are placed coaxially at distance *R* and the same current *I* flows through the loops in the same direction. The magnitude of magnetic field at *P*, midway between the centres *A* and *C*, is given by [Refer to figure given below]:



(A) $\frac{4N\mu_0 I}{5^{1/2}R}$ (B) $\frac{4N\mu_0 I}{5^{3/2}R}$ (C) $\frac{8N\mu_0 I}{5^{3/2}R}$ (D) $\frac{8N\mu_0 I}{5^{1/2}R}$

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PMG226

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	72	D	C	A	B	C	3	C	3.00	C	C

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RACE
Regular Analysis Through Continuous Exercise

Magnetic Effects of Current and Magnetism : RACE-02

1. A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is :

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) $\frac{1}{2}$ (D) 2

PMG227

2. A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r ($r < R$) from its centre will be :-

(A) $B \propto r^2$ (B) $B \propto r$ (C) $B \propto \frac{1}{r^2}$ (D) $B \propto \frac{1}{r}$

PMG228

3. A coaxial cable consists of an inner wire of radius 'a' surrounded by an outer shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) $x < a$ and (ii) $a < x < b$?

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(A) $\frac{x^2}{a^2}$ (B) $\frac{a^2}{x^2}$ (C) $\frac{x^2}{b^2-a^2}$ (D) $\frac{b^2-a^2}{x^2}$

PMG229

4. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

(A) B (B) $2B$ (C) $4B$ (D) $\frac{B}{2}$

PMG230

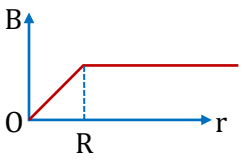
5. A proton and an alpha particle of the same enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the circular paths described by the alpha particle and proton is:

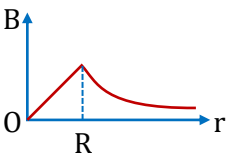
(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 2

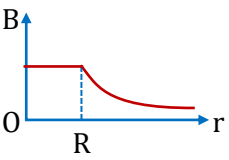
PMG231

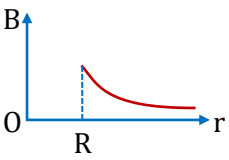
6. An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.

Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

(A) 

(B) 

(C) 

(D) 

PMG232

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7. A charge particle is moving in a uniform magnetic field $(2\hat{\imath} + 3\hat{\jmath})T$. If it has an acceleration of $(\alpha\hat{\imath} - 4\hat{\jmath})m/s^2$, then the value of α will be

(A) 3 (B) 6 (C) 12 (D) 2

PMG233

8. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an uniform magnetic field, speed and energy remains the same for a moving charged particle.

Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true.

PMG234

9. A proton, a deuteron and an α-particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

(A) $1:\sqrt{2}:\sqrt{2}$ (B) $1:1:\sqrt{2}$ (C) $\sqrt{2}:1:1$ (D) $1:\sqrt{2}:1$

PMG235

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	B	A	A	C	D	B	A	D

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RACE
Regular Analysis Through Continuous Exercise

Magnetic Effects of Current and Magnetism : RACE-03

1. A loop of flexible wire of irregular shape carrying current is placed in an external magnetic field. Identify the effect of the field on the wire.

(A) Loop assumes circular shape with its plane normal to the field.

(B) Loop assumes circular shape with its plane parallel to the field.

(C) Wire gets stretched to become straight.

(D) Shape of the loop remains unchanged.

PMG236

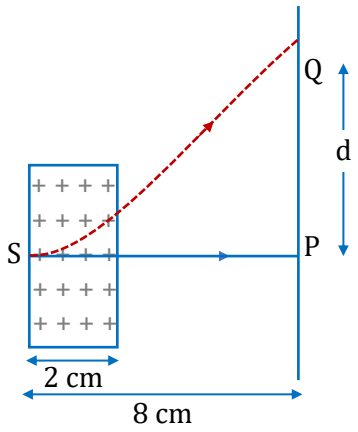
2. A deuteron and a proton moving with equal kinetic energy enter into to a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x}:1$ where x is ___.

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PMG237

3. An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3} T)\hat{k}$ at S (See figure). The field extends between x = 0 and x = 2 cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is :

(electron's charge = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg)

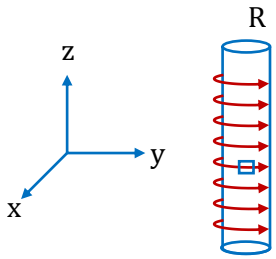


(A) 12.87 cm (B) 1.22 cm (C) 11.65 cm (D) 2.25 cm

PMG238

For More Join: @IITJEE_Advanced

4. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed v. If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning) :



(A) $\frac{e\mu_0 nIR}{m}$ (B) $\frac{e\mu_0 nIR}{2m}$ (C) $\frac{2e\mu_0 nIR}{m}$ (D) $\frac{e\mu_0 nIR}{4m}$

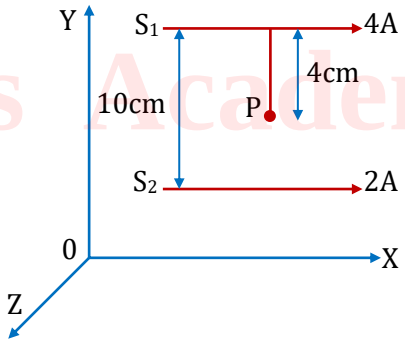
PMG239

5. Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of $4A$ and $2A$ respectively. The conductors are placed along x-axis in X-Y plane. There is a point P located between the conductors (as shown in figure).

A charge particle of 3π coulomb is passing through the point P with velocity

$\vec{v} = (2\hat{\imath} + 3\hat{\jmath})m/s$; *where* $\hat{\imath}$ & $\hat{\jmath}$ represents unit vector along x & y axis respectively.

The force acting on the charge particle is $4\pi \times 10^{-5}(-x\hat{\imath} + 2\hat{\jmath})N$. The value of x is :



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(A) 2 (B) 1 (C) 3 (D) –3

PMG240

6. Given below are two statements :

Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy: whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field. In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

PMG241

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7. Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be :

(A) 8 : 5 (B) 5 : 4 (C) 5 : 3 (D) 8 : 7

PMG242

8. A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{\jmath}$ with $B = 12\ mT$. The value E required for an electron of energy 728 eV moving along the positive x-axis to pass undeflected is :

(Given, mass of electron = $9.1 \times 10^{-31} kg$)

(A) 192 kVm^{-1} (B) 192 $m\ Vm^{-1}$ (C) 9600 kVm^{-1} (D) 16 kVm^{-1}

PMG243

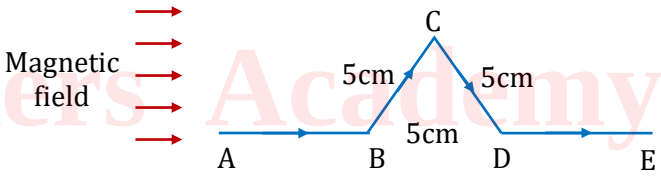
9. A cyclotron is used to accelerate protons. If the operating magnetic field is 1.0 T and the radius of the cyclotron 'dees' is 60 cm, the kinetic energy of the accelerated protons in MeV will be :

[use $m_p = 1.6 \times 10^{-27}\ kg$, $e = 1.6 \times 10^{-19}\ C$]

(A) 12 (B) 18 (C) 16 (D) 32

PMG244

10. A triangular shaped wire carrying 10A current is placed in a uniform magnetic field of 0.5T, as shown in figure. The magnetic force on segment CD is (Given $BC = CD = BD = 5\ cm$).

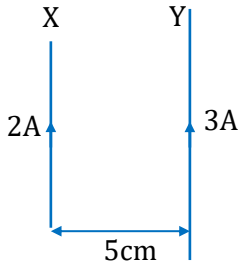


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(A) 0.126 N (B) 0.312 N (C) 0.216 N (D) 0.245 N

PMG245

11. A wire X of length 50 cm carrying a current of 2 A is placed parallel to a long wire Y of length 5 m. The wire Y carries a current of 3 A. The distance between two wires is 5 cm and currents flow in the same direction. The force acting on the wire Y is :



(A) $1.2 \times 10^{-5}\ N$ directed towards wire X. (B) $1.2 \times 10^{-4}\ N$ directed away from wire X.

(C) $1.2 \times 10^{-4}\ N$ directed towards wire X. (D) $2.4 \times 10^{-5}\ N$ directed towards wire X.

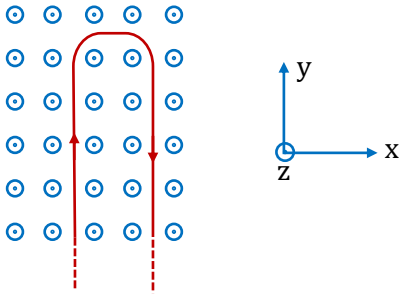
PMG246

12. Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of $10^{-5}\ N$, then separation between the wires is _____cm.

PMG247

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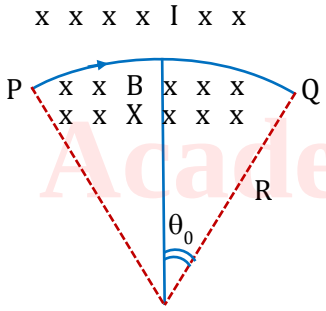
13. A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immerged in a perpendicular magnetic field $B = B_0\ \hat{j}$ as shown in figure. The magnetic force on the wire if it has a current i is :



(A) $-iBR\,\hat{j}$ (B) $2iBR\,\hat{j}$ (C) $iBR\,\hat{j}$ (D) $-2iBR\,\hat{j}$

PMG248

14. A wire carrying current I is tied between points P and Q and is in the shape of a circular arch of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by x) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the centre of the circle (of which it forms an arch) then the tension in the wire is :-



(A) IBR (B) $\frac{IBR}{2\sin\theta_0}$ (C) $\frac{IBR}{\sin\theta_0}$ (D) $\frac{IBR\theta_0}{\sin\theta_0}$

PMG249

15. Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of $x\ A$ in the same direction. If the force of attraction per meter of each wire is $2 \times 10^{-6} N$, then the value of x is approximately:

(A) 1 (B) 2.4 (C) 1.4 (D) 2

PMG250

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ANSWER KEY

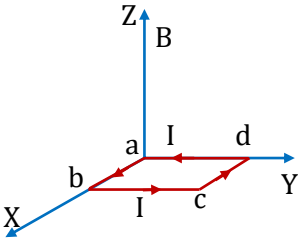
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	2	A	B	C	C	B	A	B	C
Question	11	12	13	14	15					
Answer	A	5	D	A	C					

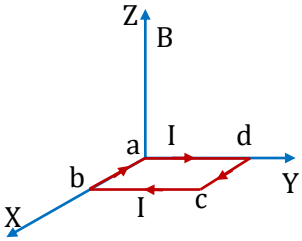
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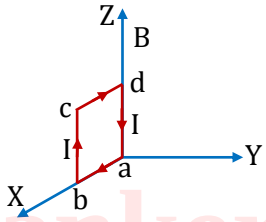
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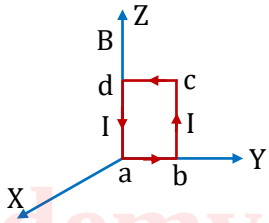
Magnetic Effects of Current and Magnetism : RACE-04

1. A uniform magnetic field B of 0.3 T is along the positive Z-direction. A rectangular loop ($abcd$) of sides 10 cm × 5 cm carries a current I of 12 A. Out of the following different orientations which one corresponds to stable equilibrium ?

(A) 

(B) 

(C) 

(D) 

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PMG251

2. A charge q is spread uniformly over an insulated loop of radius r. If it is rotated with an angular velocity ω with respect to normal axis then the magnetic moment of the loop is :-

(A) $\frac{1}{2}q\omega r^2$ (B) $q\,\omega r^2$ (C) $\frac{3}{2}q\omega r^2$ (D) $\frac{4}{3}q\omega r^2$

PMG252

3. A moving coil galvanometer has a coil with 175 turns and area 1 cm^2. It uses a torsion band of torsion constant 10^{-6} N-m/rad. The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA. The value of B (in Tesla) is approximately :-

(A) 10^{-3} (B) 10^{-1} (C) 10^{-4} (D) 10^{-2}

PMG253

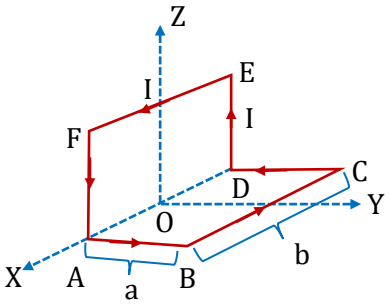
4. A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then :

(A) $T = \sqrt{\frac{\pi m}{2IB}}$ (B) $T = \sqrt{\frac{2\pi m}{IB}}$ (C) $T = \sqrt{\frac{\pi m}{IB}}$ (D) $T = \sqrt{\frac{2m}{IB}}$

PMG254

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5. A wire carrying current I is bent in the shape $ABCDEFA$ as shown, where rectangle $ABCDA$ and $ADEFA$ are perpendicular to each other. If the sides of the rectangles are of lengths a and b, then the magnitude and direction of magnetic moment of the loop $ABCDEFA$ is :



(A) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$ (B) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

(C) abI, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$ (D) abI, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

PMG255

6. A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4}\ m^2$. If a torque of 1.5 Nm is required to keep this coil parallel to magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is _____.

PMG256

7. A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm. The minimum work required to rotate it from its stable to unstable equilibrium position is :

(A) $9.2 \times 10^{-3}\ J$ (B) $6.4 \times 10^{-2}\ J$ (C) $11.7 \times 10^{-3}\ J$ (D) $7.2 \times 10^{-2}\ J$

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PMG257

8. An iron rod of volume $10^{-3}\ m^3$ and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be :

(A) $0.5 \times 10^2\ Am^2$ (B) $50 \times 10^2\ Am^2$ (C) $500 \times 10^2\ Am^2$ (D) $5 \times 10^2\ Am^2$

PMG258

9. A bar magnet having a magnetic moment of $2.0 \times 10^5\ JT^{-1}$, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5}\ T$. The work done in rotating the magnet slowly through 60° from the direction of field is :

(A) 14 J (B) 8.4 J (C) 4 J (D) 1.4 J

PMG259

10. The susceptibility of a paramagnetic material is 99. The permeability of the material in $Wb/A\text{-}m$ is : [Permeability of free space $\mu_0 = 4\pi \times 10^{-7} Wb/A - m$]

(A) $4\pi \times 10^{-7}$ (B) $4\pi \times 10^{-4}$ (C) $4\pi \times 10^{-5}$ (D) $4\pi \times 10^{-6}$

PMG260

11. The value of aluminium susceptibility is 2.2×10^{-5}. The percentage increase in the magnetic field if space within a current carrying toroid is filled with aluminium is $\frac{x}{10^4}$. Then the value of x is _____.

PMG261

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12. Given below are two statements:

Statement-I: Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.

Statement-II: Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.

Choose the **CORRECT** answer from the options given below : -

(A) Both statement – I and statement –II are true.

(B) Both statement – I and Statement – II are false.

(C) Statement – I is true but statement – II is false.

(D) Statement-I is false but Statement-II is true.

PMG262

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	B	A	20	D	D	A	C
Question	11	12								
Answer	22	A								

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